

U Hari Charan

Palamaner, Andhra Pradesh

+91 9849136728 — haricharan45u@gmail.com

GitHub — LinkedIn

Education

VIT-AP University

Graduation: 2028

Bachelor of Technology in Computer Science and Engineering

CGPA: **8.4/10**

Sri Chaitanya Junior College

Intermediate (Class XII): **95.5%**

Elena Bettini School

SSC (Class X): **91%**

Technical Skills

Programming Languages: Java, Python, C++, C, JavaScript

Frameworks: React, Flutter, FastAPI, Express.js, HTML, CSS

Databases: MongoDB, PostgreSQL, Firebase

Developer Tools: Git, GitHub, Docker, VS Code

Core Subjects: Data Structures and Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Software Engineering

Domains: Artificial Intelligence, Machine Learning, Computer Vision, Cybersecurity, Full Stack Development, REST APIs

Projects

SecureX

SecureX is a cybersecurity platform developed to detect and mitigate security threats through secure backend services and authentication mechanisms. The application integrates REST APIs with MongoDB and Docker to provide a scalable architecture while ensuring secure communication and efficient data management.

Technologies: Java, FastAPI, MongoDB, Docker, Git

BDH Healthcare

BDH Healthcare is an AI-powered healthcare platform designed for adverse event learning and clinical decision support. The system integrates intelligent AI workflows with scalable backend services to process healthcare data efficiently and generate meaningful insights for medical applications.

Technologies: Python, FastAPI, AI/ML, MongoDB

Civic Pulse

Civic Pulse is a civic issue reporting platform that enables citizens to submit and track public complaints through an intuitive interface. It offers RESTful APIs, database integration, and efficient complaint management to improve communication between citizens and authorities.

Technologies: React, Express.js, MongoDB, Node.js

AI Crop Monitoring

Developed an AI-based agricultural monitoring system using satellite imagery and machine learning techniques to analyze crop health and environmental conditions. The project applies GIS concepts and predictive analytics to support precision agriculture and improve farming decisions.

Technologies: Python, Machine Learning, GIS, Remote Sensing

GridShield

GridShield is a cybersecurity solution for industrial control systems that monitors network traffic and identifies potential cyber threats. The project focuses on anomaly detection, secure communication, and protection of critical infrastructure.

Technologies: Python, Networking, Cybersecurity, Docker

Research Publication

Human-Following Robot: Smart Assistance Using Ultrasonic and IR Sensors

Co-authored a research paper titled *Human-Following Robot: Smart Assistance Using Ultrasonic and IR Sensors*, presented at the **International Conference on Advances, Architectures, Algorithms and Applications for Artificial Intelligence (IC4AI 2026)**, VIT-AP University, Amaravati. The research proposes a low-cost autonomous human-following robot that combines ultrasonic and infrared sensors for real-time human tracking, obstacle avoidance, and smart assistance. The work also discusses modular sensing architecture, sensor fusion, navigation strategies, and future integration with vision-based perception and SLAM for intelligent robotic applications.

Achievements

- Top 10 Team at Postathon Hackathon conducted by VIT-AP University.
- Finalist at Synaptic Hackathon conducted by IIT Madras.
- Co-authored and presented a research paper at IC4AI 2026, VIT-AP University.
- Developed multiple AI, healthcare, cybersecurity, and full-stack projects through hackathons and academic initiatives.

Relevant Coursework

Data Structures and Algorithms • Database Management Systems • Operating Systems • Computer Networks • Artificial Intelligence • Machine Learning • Software Engineering

Coding Profiles

GitHub: github.com/Hari4507

LinkedIn: [linkedin.com/in/hari-charan-u-94aa71302](https://www.linkedin.com/in/hari-charan-u-94aa71302)

Areas of Interest

Artificial Intelligence • Machine Learning • Software Engineering • Cybersecurity • Full Stack Development • Robotics • Hackathons